

Hull Chapter 2 - Complete Mastery Guide

Mechanics of Futures Markets - 9th Global Edition

Study Guide

Chapter 2 - Complete Mastery Guide

This guide follows the concepts in **Chapter 2, “Mechanics of Futures Markets,”** of John C. Hull’s *Options, Futures, and Other Derivatives, 9th Global Edition*. The goal is not only to recognize the definitions, but to be able to answer the chapter’s conceptual and calculation questions confidently.

The whole chapter can be organized around one central question:

How does a futures exchange turn a promise about a future transaction into a standardized, liquid contract that can be traded safely every day?

Chapter 1 told us what a futures contract is. Chapter 2 explains the machinery that makes a futures market actually work.

The most important ideas are:

- how a futures position is created and closed;
 - how the exchange specifies the contract;
 - why futures and spot prices converge at delivery;
 - how margin and daily settlement control default risk;
 - how the clearing house sits behind the market;
 - how OTC markets borrow the same ideas through CCPs and collateral;
 - how to read futures quotes;
 - what volume and open interest mean;
 - how delivery works;
 - the main types of traders and orders;
 - regulation, accounting, and tax treatment in the chapter’s framework;
 - and the practical differences between forwards and futures.
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1. Futures recap: what exactly are we trading?

A **futures contract** is an agreement to buy or sell an underlying asset at a future time at a price determined today.

The buyer is **long**.

The seller is **short**.

If the futures price rises after the trade:

- the long gains;
- the short loses.

If the futures price falls:

- the long loses;
- the short gains.

At the simplest level, for a change in futures price from F_0 to F_1 on a position representing Q units,

$$\text{Gain to long} = Q(F_1 - F_0)$$

and

$$\text{Gain to short} = Q(F_0 - F_1).$$

The crucial Chapter 2 difference from a forward contract is that a futures contract is **standardized, exchange-traded, cleared, margined, and settled day by day**.

2. How a futures contract comes into existence

Imagine Trader A wants to buy September corn futures and Trader B wants to sell September corn futures.

Trader A's instruction is effectively:

“Take a long position in one September corn futures contract.”

Trader B's instruction is:

“Take a short position in one September corn futures contract.”

When compatible buy and sell orders meet, a trade occurs at an agreed futures price.

If one corn contract represents 5,000 bushels and the futures price is 600 cents per bushel, then the economic exposure represented by the contract is

$$5,000 \times \$6.00 = \$30,000.$$

That does **not** mean the trader pays \$30,000 to enter the futures contract. Instead, the trader posts margin, discussed later.

The price itself is determined by supply and demand in the futures market. If buying pressure dominates, the futures price tends to rise; if selling pressure dominates, it tends to fall.

3. Closing out a futures position

Most futures contracts never result in delivery.

A trader normally exits by taking the **opposite position in the same contract**.

If you originally bought one September corn futures contract, you close it by selling one September corn futures contract.

If you originally sold one December gold futures contract, you close it by buying one December gold futures contract.

So:

Close long $\Rightarrow$ sell the same futures contract

Close short $\Rightarrow$ buy the same futures contract

You do not have to locate the original person on the other side of your trade. The clearing system makes the identities of the original counterparties irrelevant.

Example

You buy a futures contract at

$$F_0 = 100.$$

Later you close it by selling at

$$F_1 = 107.$$

For Q units,

$$\text{Profit} = Q(107 - 100) = 7Q.$$

The market does not care who was originally short against you.

4. Why delivery still matters if almost nobody delivers

This is a subtle but fundamental idea.

Most traders close out before delivery, yet the **possibility of delivery** is what anchors the futures price to the underlying spot market.

If delivery were impossible, the futures price could drift away from the economic value of the actual asset with no mechanism forcing the two back together.

Because someone can ultimately deliver or receive the real asset, futures and spot prices must become very close as maturity approaches.

This is the foundation of **convergence**, discussed in Section 2.3 of the book.

5. Contract specification: why standardization matters

For an exchange-traded contract to be liquid, every trader must know exactly what one contract means.

The exchange therefore specifies:

1. the underlying asset;
2. acceptable quality or grade;
3. contract size;
4. delivery location;
5. delivery month and delivery period;
6. how prices are quoted;
7. trading hours and final trading rules;
8. daily price limits where applicable;
9. position limits where applicable.

Standardization allows thousands of strangers to trade the same contract without renegotiating its legal details each time.

6. Specification of the underlying asset

For a financial asset such as a currency, specification is usually straightforward. One Japanese yen is economically the same asset as another Japanese yen.

For commodities, quality can vary.

Corn, wheat, oil, cattle, orange juice, and metals can exist in different grades or qualities.

The exchange therefore has to answer:

“What exactly counts as acceptable delivery?”

Sometimes one grade is the standard, while alternative grades can be delivered with a price adjustment.

This prevents the short from satisfying a contract by delivering an inferior product that the long never intended to accept.

Exam intuition

If the asset quality is **poorly specified**, the contract becomes unattractive because buyers do not know what they may receive. Liquidity can fall and disputes can increase.

7. Delivery choices usually belong to the short

When the contract allows alternative grades, locations, or sometimes dates, the **short** often chooses among the permitted alternatives.

Why?

Because the short is the party responsible for making delivery.

This optionality has value to the short. If several deliverable assets are allowed, the short will generally choose the cheapest or most convenient acceptable alternative.

A useful intuition is:

More delivery flexibility for the short tends to favor the short

That is why questions may ask whether delivery options increase or decrease the futures price. Other things equal, giving valuable choices to the short can make the short side more attractive, which tends to put downward pressure on the contract price relative to an otherwise identical contract without those choices.

8. Contract size

The **contract size** says how much underlying one futures contract represents.

Examples in the chapter include contracts representing quantities such as:

- 5,000 bushels of grain;
- 100 ounces of gold;
- 1,000 barrels of crude oil;
- 40,000 pounds of live cattle.

Contract size is a design trade-off.

If it is too large, smaller hedgers and speculators cannot use the contract easily.

If it is too small, a large position requires many contracts, increasing transaction and operational costs.

This is why exchanges sometimes create **mini** or smaller contracts.

9. Number of contracts needed

A very common calculation is:

$$N = \frac{\text{Exposure size}}{\text{Contract size}}$$

Suppose a cattle farmer will sell 120,000 pounds of cattle and each futures contract covers 40,000 pounds.

Then

$$N = \frac{120,000}{40,000} = 3.$$

Because the farmer will **sell** cattle later and fears a price decline, the natural hedge is to **short 3 futures contracts**. This logic connects Chapter 2 mechanics with the hedging ideas developed further in Chapter 3.

10. Delivery location

For physical commodities, location matters because transportation costs matter.

A ton of a commodity in one city is not economically identical to a ton located thousands of kilometers away.

The exchange specifies valid delivery locations.

If several locations are allowed, the contract may include location-based price adjustments.

The important idea is that contract design must eliminate ambiguity about:

what, where, and when

is delivered.

11. Delivery month and delivery period

A futures contract is usually identified by its **delivery month**.

For example:

December gold futures

The exchange specifies:

- when trading in that maturity begins;
- the delivery period;
- the first day notices may be submitted;
- the final day notices may be submitted;
- the final trading day.

Delivery is often possible during a range of days rather than on one exact date.

This is one reason futures differ from many forward contracts, which commonly specify one particular settlement date.

12. Price quotation conventions

The exchange decides how a futures price is quoted.

Different contracts use different units.

Examples can include:

- dollars per ounce;
- cents per pound;
- cents per bushel;
- dollars per barrel;
- points times a contract multiplier;

- bond prices quoted in fractions.

Before doing any numerical futures problem, identify:

$$\boxed{\text{quoted unit} \times \text{contract size}}$$

Otherwise it is easy to be wrong by factors of 100 or 1,000.

13. Daily price limits

Some futures contracts have rules limiting how far prices can move in one trading day.

If the price reaches the maximum permitted increase, the contract may be described as **limit up**.

If it reaches the maximum permitted decrease, it may be **limit down**.

The purpose is to reduce extreme short-term movements and disorderly trading.

But price limits can also create a problem: if the economic equilibrium price is far away, the market may be unable to reach it immediately.

So price limits are not unambiguously beneficial.

14. Position limits

A **position limit** is a cap on the number of contracts a trader or group may hold in certain circumstances.

The aim is to reduce the risk that a speculator acquires enough market power to distort prices or manipulate delivery.

Position limits are especially relevant to the chapter's later discussion of attempts to **corner a market**.

15. Convergence of futures price to spot price

This is one of the most important concepts in the chapter.

Let

$$F_T = \text{futures price at delivery}$$

and

$$S_T = \text{spot price at delivery.}$$

As the delivery period is reached,

$$\boxed{F_T \approx S_T}$$

and, ideally,

$$\boxed{F_T = S_T}$$

apart from small market frictions.

Why must this happen?

Because otherwise traders have an arbitrage or a delivery-based economic advantage.

16. If futures price is above spot during delivery

Suppose during the delivery period:

$$F_T > S_T.$$

A trader can conceptually:

1. buy the asset in the spot market at S_T ;
2. short the futures at F_T ;
3. deliver the asset into the futures contract.

Ignoring transaction costs,

$$\text{Profit} = F_T - S_T > 0.$$

This creates selling pressure in futures and buying pressure in spot.

As arbitrageurs act:

- futures price tends to fall;
 - spot price may tend to rise;
 - the gap disappears.
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17. If futures price is below spot during delivery

Suppose instead:

$$F_T < S_T.$$

A company that genuinely wants the underlying asset can buy futures and take delivery rather than purchasing the asset at the more expensive spot price.

This increases demand for the futures contract and tends to push the futures price upward.

In idealized settings where the asset can also be shorted conveniently, one can express the arbitrage more symmetrically. But Hull's Chapter 2 explanation emphasizes the practical acquisition route because short-selling a physical commodity may be difficult.

The important conclusion is:

Delivery prevents a persistent gap between futures and spot at maturity

18. Margin: the central safety mechanism

Imagine a futures market with no margin.

You agree today to buy an asset later. The market moves badly against you. Your loss becomes huge. At maturity you simply say:

“I cannot pay.”

The other side now has a credit problem.

Futures exchanges reduce this risk through:

- initial margin;
- daily settlement;
- maintenance margin;
- margin calls;
- variation margin;
- clearing-house guarantees and guaranty funds.

This is the operational heart of Chapter 2.

19. Initial margin

The **initial margin** is the amount deposited when the futures position is opened.

It is not the purchase price of the underlying asset.

It is better thought of as a **performance bond** or security deposit.

Suppose two gold futures contracts represent 200 ounces in total and the futures price is

$$\$1,450/\text{oz.}$$

The economic exposure is

$$200 \times 1,450 = \$290,000.$$

But the required initial margin might be only, for example,

$$\$12,000.$$

This is one source of futures leverage.

20. Marking to market / daily settlement

Futures gains and losses are realized **every trading day**.

This is called:

- **marking to market**, or
- **daily settlement**.

Suppose you are long 200 ounces of gold futures.

Yesterday's settlement price:

$$1,450.$$

Today's settlement price:

$$1,441.$$

Price change:

$$1,441 - 1,450 = -9.$$

Long's daily gain:

$$200(-9) = -\$1,800.$$

So \$1,800 is removed from the long's margin account and transferred through the clearing system to the short side. If instead the price rose by \$9, the long would gain \$1,800.

21. Daily gain formulas

For a **long futures position** representing Q units:

$$\text{Daily gain} = Q(F_t - F_{t-1})$$

For a **short futures position**:

$$\text{Daily gain} = Q(F_{t-1} - F_t)$$

Over several days, the total gain is the sum of daily gains.

If the position is entered at F_0 and closed at F_n , ignoring interest on intermediate cash flows:

$$\sum_{t=1}^n Q(F_t - F_{t-1}) = Q(F_n - F_0)$$

for a long.

This telescoping identity explains why total futures profit resembles forward profit even though the **timing** of cash flows differs.

22. Why daily settlement reduces credit risk

Suppose a contract accumulates a \$500,000 loss over six months.

With a forward, much of that loss may remain unpaid until maturity.

With a futures contract, losses are transferred day by day.

That means the system does not allow a giant unpaid liability to quietly accumulate for months.

Conceptually:

$$\text{Frequent settlement reduces the amount that can remain unpaid}$$

This is a major reason futures exchanges have historically been able to manage counterparty risk effectively.

23. Maintenance margin

The **maintenance margin** is a threshold below the initial margin.

Example:

Initial margin:

$$12,000$$

Maintenance margin:

$$9,000$$

If losses reduce the account **below** \$9,000, the trader receives a **margin call**.

Important:

The trader usually does not merely restore the balance to \$9,000.

The account is normally restored to the **initial margin level**, here \$12,000.

24. Variation margin

The extra amount deposited after a margin call is called **variation margin** in the chapter's margin-account discussion.

Suppose the balance falls to

$$7,980.$$

Initial margin is

$$12,000.$$

Required top-up:

$$12,000 - 7,980 = \boxed{4,020}.$$

That brings the account back to the initial margin level.

25. What if the trader ignores a margin call?

The broker will normally close out the position.

If you are long, the broker sells offsetting futures.

If you are short, the broker buys offsetting futures.

This prevents the broker and clearing system from allowing an undercollateralized losing position to continue indefinitely.

The trader can still owe money if losses exceed available margin.

26. Margin-call price-change formula

This is one of the most useful Chapter 2 formulas.

Let:

- M_0 = initial margin;
- M_m = maintenance margin;
- Q = total number of underlying units represented by the position.

The loss that can occur before the balance reaches maintenance is approximately

$$M_0 - M_m.$$

Therefore the adverse futures-price move corresponding to that amount is

$$\Delta F = \frac{M_0 - M_m}{Q}$$

For a **long**, the adverse move is a fall.

For a **short**, the adverse move is a rise.

Strictly, the margin call occurs once the balance moves **below** the maintenance threshold, so the exact trigger is just beyond this amount.

27. Margin-call example: short silver

Suppose you short one silver futures contract.

Contract size:

$$5,000 \text{ oz}$$

Initial margin:

$$4,000$$

Maintenance margin:

$$3,000.$$

You can absorb approximately

$$4,000 - 3,000 = 1,000$$

of loss before reaching maintenance.

Price rise causing \$1,000 loss:

$$\frac{1,000}{5,000} = \$0.20/\text{oz}.$$

Because you are **short**, a price **increase** hurts you.

If the initial futures price is \$17.20, the account reaches maintenance around

$$17.20 + 0.20 = 17.40.$$

A move slightly beyond that causes a margin call.

This is exactly the logic required for the chapter's margin questions.

28. When can money be withdrawn from a margin account?

If gains push the balance **above the initial margin**, the excess can generally be withdrawn.

Suppose initial margin is

$$12,000$$

and the account rises to

$$14,500.$$

Then the excess is

$$14,500 - 12,000 = 2,500.$$

So up to \$2,500 can be withdrawn, subject to broker rules.

A useful calculation is therefore:

$$\text{Withdrawable excess} = \max(\text{Margin balance} - M_0, 0)$$

29. Securities and haircuts

Initial margin can sometimes be satisfied partly with approved securities rather than only cash.

However, a security worth \$100 in the market may be credited for less than \$100 for margin purposes.

The reduction is called a **haircut**.

If a security worth \$100 has a 10% haircut, its collateral value is

$$100(1 - 0.10) = 90.$$

Haircuts protect the clearing system against the possibility that the collateral itself loses value before it can be sold.

30. Initial margin is not the same as economic cost

Do not confuse **cash posted as margin** with a purchase price.

Margin remains the trader's asset, subject to gains/losses and the rules of the account.

Brokers may pay interest on margin balances, and approved securities may sometimes be used for initial margin.

So if a futures position has \$300,000 of notional exposure and requires \$15,000 initial margin, it does not mean the trader "bought \$300,000 for \$15,000."

It means \$15,000 of collateral supports a much larger market exposure.

That distinction is essential for understanding leverage correctly.

31. Long and short futures have symmetric margin requirements

A useful feature of futures is that taking a short position is operationally almost as easy as taking a long position.

Both sides post margin.

This differs from the spot market, where:

- buying an asset is straightforward;
- short-selling an asset can require borrowing it and may face restrictions.

Futures therefore make directional long and short exposure relatively symmetric.

32. Clearing house: the system behind the futures market

The clearing house acts as the central intermediary.

Instead of Trader A worrying directly about Trader B's ability to pay, the market structure is effectively:

$$\text{Trader A} \leftrightarrow \boxed{\text{Clearing House}} \leftrightarrow \text{Trader B.}$$

The clearing house:

- records and nets positions;
- collects margin;
- transfers daily gains and losses;
- monitors members;
- manages default procedures;
- guarantees performance subject to its rulebook and resources.

33. Clearing members and brokers

Not every broker is necessarily a direct clearing-house member.

A broker that is not a clearing member routes its futures business through a clearing member.

So the chain can look like:

$$\text{Client} \rightarrow \text{Broker} \rightarrow \text{Clearing Member} \rightarrow \text{Clearing House.}$$

The client maintains margin with the broker.

The clearing member maintains margin with the clearing house.

These are related but distinct accounts.

34. Broker margin vs clearing-house margin

This distinction is specifically tested in Chapter 2.

A broker may impose margin requirements on its client that are **higher** than the exchange minimum.

The client's account typically has:

- initial margin;
- maintenance margin;
- margin calls when balance drops below maintenance.

For the clearing member's account with the clearing house, the structure can differ. In Hull's description, the clearing member provides clearing margin and settles variation margin daily; the same retail-style maintenance-margin mechanism is not applied in exactly the same way.

The clearing house focuses on the member's overall cleared obligations, often with netting of long and short positions under the applicable rules.

35. Guaranty fund

Clearing members contribute to a **guaranty fund** or default fund.

If a member fails and its margin is insufficient to cover losses when positions are closed, the fund provides an additional layer of financial protection.

A simplified safety waterfall is:

1. losing trader's margin;
2. broker/clearing member resources under the rules;
3. default or guaranty fund resources;
4. other clearing-house protections.

The exact modern waterfall varies by clearing house, but the Chapter 2 principle is that mutualized resources help protect the integrity of the market.

36. OTC markets: why Chapter 2 returns to them

Chapter 1 introduced OTC derivatives. Chapter 2 explains how OTC markets adopted exchange-like mechanisms after the financial crisis.

Traditionally, an OTC derivative could leave each party exposed to the other's default.

If A owes B a large positive amount and A defaults, B may suffer a loss.

Two major approaches reduce this risk:

1. **central clearing through a CCP;**
 2. **bilateral clearing with collateral agreements.**
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37. CCP - Central Counterparty

A **CCP** for OTC derivatives performs a role similar to a futures clearing house.

Suppose A and B agree on a standardized OTC derivative.

After central clearing, the economic structure becomes:

$$A \leftrightarrow CCP \leftrightarrow B.$$

The CCP becomes counterparty to both sides.

Members typically provide:

- initial margin;
- variation margin;
- default-fund contributions.

This reduces direct bilateral counterparty exposure and centralizes risk management.

38. Bilateral clearing

If an OTC transaction is not centrally cleared, it may be handled **bilaterally**.

The two parties commonly operate under a master agreement covering many transactions.

A **Credit Support Annex (CSA)** can define collateral rules.

If the portfolio moves in favor of A, B may have to post collateral to A.

If it moves in favor of B, A may have to post collateral to B.

This resembles variation margin economically, even though the legal and settlement mechanics differ from exchange futures.

39. Variation margin vs collateral in OTC markets

The conceptual similarity is:

losing side posts value; winning side receives protection

But there is an important distinction in Hull's discussion.

In exchange-traded futures, daily variation margin is part of **daily settlement**.

In many OTC arrangements, variation collateral secures exposure but the underlying transaction is not necessarily legally settled and reset in exactly the same way each day.

This is why the treatment of interest on cash variation amounts can differ.

40. Initial margin in bilateral OTC clearing

Historically, bilateral agreements often focused primarily on variation collateral.

Post-crisis regulation increased the use of **initial margin** for many uncleared transactions between financial institutions.

Initial margin protects against the potential change in exposure during the period between a counterparty default and the time the position can be replaced or closed.

This is conceptually different from variation margin:

- **variation margin** covers exposure that has already appeared;
- **initial margin** protects against potential future exposure during close-out.

41. Why collateral reduces credit risk but does not eliminate all risk

Suppose two institutions post collateral daily.

That makes a default less damaging because much of the current exposure is already secured.

But collateral does not remove:

- market risk;
- liquidity risk;
- model risk;
- gap risk between margin calls;
- operational risk;
- the risk that collateral itself falls in value;
- concentration risk.

Hull's LTCM example is useful because LTCM collateralized its trades, yet extreme market moves plus enormous leverage created a severe liquidity problem.

The lesson is:

Collateral controls counterparty exposure, not every form of risk

42. Bilateral clearing vs central clearing

Bilateral clearing	Central clearing
A deals directly with B	CCP stands between A and B
Many pairwise legal relationships	Centralized clearing relationship
Credit exposure depends on each counterparty agreement	CCP risk-management framework applies
Collateral defined by CSA/master agreement	Margin defined by CCP rules
Default management is bilateral	Default management centralized
Netting mainly within each bilateral relationship	Multilateral netting may be possible through CCP structure

Central clearing can reduce and simplify certain counterparty exposures, but it also concentrates risk management in the CCP. A CCP must therefore be extremely robust.

43. Reading futures market quotes

A futures quote table often includes fields such as:

- contract month;
- open;
- high;
- low;
- prior settlement;
- last trade;
- change;
- volume;
- sometimes open interest.

You should understand each rather than treating the table as random numbers.

44. Open, high, low, last

Open: a representative price near the start of the trading session.

High: highest traded price so far during the session.

Low: lowest traded price so far during the session.

Last: the most recent transaction price.

These are trading statistics.

They are not necessarily the same as the official settlement price.

45. Settlement price

The **settlement price** is the exchange-defined price used for daily margin calculations.

It is often based on trading near the close, but the exact method is determined by the exchange.

This price matters because daily gains and losses are based on settlement prices.

Suppose yesterday's settlement is \$1,434.30 and today's settlement is \$1,425.30 for a 100-ounce gold contract.

For one long contract:

$$100(1,425.30 - 1,434.30) = -\$900.$$

The short gains \$900.

46. Trading volume

Trading volume is the number of futures contracts traded during a period, usually a day.

If one buyer and one seller execute one contract, that counts as **one contract of volume**, not two.

Volume measures trading activity.

It does **not** measure the number of contracts currently outstanding.

47. Open interest

Open interest is the number of futures contracts that remain outstanding.

Equivalently, it is:

- the number of open long contracts;
- which must equal the number of open short contracts.

If open interest is 10,000:

$$10,000 \text{ long contracts} = 10,000 \text{ short contracts.}$$

Do **not** add them and say 20,000 contracts. Every contract has one long and one short side.

48. Volume vs open interest

This distinction is directly tested.

Volume	Open interest
How much trading occurred today	How many contracts remain open
Flow measure	Stock measure
Can be very large because the same contract exposure can turn over repeatedly	Changes only when positions are created or extinguished

Day traders can generate very high volume while ending the day with little or no open position.

49. How a trade changes open interest

There are four important cases.

Case 1: New long trades with new short

A new contract is created.

$$\Delta OI = +1$$

Case 2: Existing long closes against existing short closing

A contract disappears.

$$\Delta OI = -1$$

Case 3: New long buys from an existing long who is closing

One position replaces another.

$$\Delta OI = 0$$

Case 4: Existing short closes by buying from a new short seller

Again, one open position is transferred.

$$\Delta OI = 0$$

This is why a single futures trade can increase, decrease, or leave open interest unchanged.

50. Fast open-interest formula from participant counts

If buyer data are classified as:

- new longs;
- buyers closing shorts;

and seller data are:

- new shorts;
- sellers closing longs,

then

$$\Delta OI = \text{new longs} - \text{longs closed}$$

which must also equal

$$\Delta OI = \text{new shorts} - \text{shorts closed}.$$

For example, if 600 buyers are new longs and 1,200 sellers are closing existing longs,

$$\Delta OI = 600 - 1,200 = -600.$$

That is the style of reasoning required in the chapter's open-interest problem.

51. Normal market and inverted market

Hull uses two descriptive shapes for futures prices across maturities.

A **normal market** in the chapter's terminology is one where futures prices increase with maturity.

An **inverted market** is one where futures prices decrease with maturity.

You may also hear the terms **contango** and **backwardation** in practice, but Hull warns that these terms have more precise economic interpretations later in the book.

For Chapter 2, the key skill is simply reading the maturity curve correctly.

52. Delivery procedure

Even though delivery is rare, you should know the sequence.

The short decides to deliver and submits a **notice of intention to deliver** through the appropriate clearing process.

The notice specifies relevant contract details such as:

- number of contracts;
- grade, when choices exist;
- location, when choices exist.

The exchange assigns delivery to a long position according to its rules.

The person who eventually receives delivery is not necessarily the trader who originally took the opposite side of the short's trade.

This is another consequence of clearing and position transfer.

53. Warehouse receipt

For many physical commodities, delivery does not mean a truck literally arrives at the long trader's office.

Instead, delivery can involve transfer of a **warehouse receipt** representing ownership of the stored commodity.

The long then becomes responsible for relevant storage costs.

This is an important practical distinction between financial ownership and physical movement.

54. Three critical delivery-related dates

You should know these clearly.

First notice day: the first day on which a short may submit a notice of intention to deliver.

Last notice day: the final day on which such a notice may be submitted.

Last trading day: the final day on which the futures contract can be traded.

For a long trader who absolutely does not want delivery, the safest operational principle is:

close the position before first notice day

subject to the specific contract's rules.

55. Cash settlement

Some futures contracts are not conveniently deliverable.

For a stock index, for example, physical delivery would require transferring a large portfolio of constituent shares. Instead the contract can be **cash settled**.

At final settlement, the contract is valued using a specified index or spot value and gains/losses are paid in cash. No physical underlying needs to change hands.

56. FCMs and locals

Hull distinguishes two execution-side trader categories.

A **Futures Commission Merchant (FCM)** handles futures business for clients and earns commissions/fees.

A **local** trades for their own account.

In older floor-trading terminology, locals were physically present exchange traders trading their own capital.

The modern market is more electronic, but the conceptual distinction between **agency/client business** and **proprietary trading** remains useful.

57. Scalpers, day traders, and position traders

Speculators can also be categorized by holding period.

Scalper: seeks very small short-term price moves; positions may last minutes.

Day trader: closes positions before the end of the trading day and avoids overnight exposure.

Position trader: holds for longer periods, hoping to profit from larger market movements.

The categories differ mainly in time horizon, not in the fundamental long/short mechanics.

58. Market order

A **market order** says:

Execute immediately at the best available market price.

Advantage:

- high probability of execution.

Disadvantage:

- the exact price is not guaranteed.

In a fast market, the actual execution can be meaningfully different from the last price you saw.

59. Limit order

A **limit order** controls price rather than guaranteeing execution.

For a buy limit at \$30:

buy at 30 or lower

For a sell limit at \$30:

sell at 30 or higher

The trade-off is:

better price control, but the order may never execute.

60. Stop order / stop-loss order

A **stop order** becomes a market order once the stop level is reached or crossed according to the market's trigger rules.

Example:

Current price:

35

You are long and place a stop-sell at

30.

If the market falls to the stop trigger, the order becomes a market order to sell.

Purpose:

limit further loss if the market moves adversely

Important: a stop price is **not a guaranteed execution price**. Once triggered, the order executes at the best available market price, which may be worse in a fast-moving market.

61. Limit order vs stop order

This is a favorite conceptual distinction.

Suppose current price is \$2.

A **limit order to sell at \$2** means:

sell at \$2 or higher; do not accept a worse price.

A **stop order to sell at \$2** means:

if the market falls to the stop level, activate a market sell order.

So:

- limit order seeks a sufficiently **favorable execution price**;
 - stop order is usually designed to react to an **unfavorable price movement**.
-

62. Stop-limit order

A **stop-limit order** combines a trigger and a price constraint.

Example:

Stop-sell:

20.30

Limit:

20.10.

When the stop is triggered, the order becomes a limit sell order with the specified limit.

This can protect against an extremely bad execution price, but creates a new risk:

the market may gap through the limit and the order may not execute at all.

So a stop-limit sacrifices execution certainty in exchange for price control.

63. Market-if-touched order

A **market-if-touched (MIT)** order becomes a market order after the market reaches a specified favorable level.

Hull contrasts it with a stop order.

For a trader who is long:

- stop-sell below the market is typically used to limit a loss;
- MIT sell above or at a favorable target can be used to take a profit once the target is reached.

Think:

Stop: react to bad movement

MIT: react to sufficiently good movement

64. Discretionary / market-not-held order

A **discretionary** or **market-not-held** order gives the broker some discretion over execution timing in an attempt to obtain a better price.

It is not simply an instruction to execute instantly at any available price.

The broker may delay execution within the permitted discretion.

65. Time conditions on orders

Hull also introduces time instructions.

Day order: valid only for the current trading day unless executed.

Time-of-day order: valid during a specified time interval.

Good-till-canceled (GTC) / open order: remains active until executed, canceled, or until the contract's relevant trading termination under the broker/exchange rules.

Fill-or-kill (FOK): execute immediately in accordance with its terms or cancel immediately.

The key is to separate two dimensions:

1. **price/execution condition** - market, limit, stop, MIT, etc.;

2. **time-in-force condition** - day, GTC, FOK, etc.
-

66. Regulation: what Chapter 2 wants you to know

The chapter discusses US futures regulation, particularly the role of the **Commodity Futures Trading Commission (CFTC)** and the **National Futures Association (NFA)**.

Conceptually, regulation aims to support:

- market integrity;
- public price reporting;
- position reporting;
- licensing and oversight;
- customer protection;
- disciplinary procedures;
- anti-manipulation rules.

The chapter also notes the expansion of derivatives regulation after the financial crisis, including central clearing and organized trading requirements for many standardized OTC derivatives.

Important date caveat: the detailed regulatory descriptions in a 9th-edition textbook reflect the period in which that edition was written. For an exam on the book, learn the chapter's framework. For current legal compliance, always check current rules.

67. Cornering the market

A **corner** is an attempt by a trader or group to gain enough control over futures positions and/or deliverable supply to distort the market.

A simplified pattern is:

1. accumulate a very large long futures position;
2. gain influence over physical supply;
3. shorts approach delivery and struggle to obtain the commodity;
4. shorts desperately buy futures or physical supply;
5. prices can spike.

Regulators and exchanges may respond with:

- higher margin;
 - tighter position limits;
 - restrictions on new positions;
 - forced reductions or close-outs under applicable rules.
-

68. Front running

Front running occurs when a trader improperly uses knowledge of a customer's pending order to trade first for their own benefit.

Example:

A broker knows a customer is about to submit a very large buy order likely to push price upward.

The broker buys first for their own account, then the customer order moves the price higher.

This is an abuse of confidential order information and is prohibited under applicable market rules.

69. Accounting treatment: core intuition

Hull's Chapter 2 distinguishes ordinary mark-to-market recognition from **hedge accounting**.

Without hedge accounting, changes in the derivative's market value are generally recognized as they occur under the framework described in the chapter.

If the derivative qualifies as an accounting hedge, gains or losses may be recognized in a way that matches the timing of the item being hedged.

Why?

Because otherwise the accounts might show a futures gain in one reporting period and the economically offsetting physical-market loss in another, making the hedge appear artificially volatile.

The principle is:

Hedge accounting tries to match hedge effects with the hedged item

70. Accounting example

Suppose a company buys corn futures in September.

Futures price when entered:

650 cents/bushel

Year-end futures price:

670

Close-out price next year:

680.

Contract size:

5,000 bushels.

Economic total futures gain:

$$5,000(6.80 - 6.50) = \$1,500.$$

If not treated as a qualifying hedge in the book's example, the gain is split by reporting period:

Year 1:

$$5,000(6.70 - 6.50) = \$1,000$$

Year 2:

$$5,000(6.80 - 6.70) = \$500.$$

If it qualifies for hedge accounting and hedges a purchase occurring in Year 2, the accounting recognition is matched to the hedged transaction under the relevant accounting treatment.

71. Tax treatment in the chapter

Hull discusses US tax rules as they existed for the textbook's context, including distinctions between:

- ordinary income;
- capital gains/losses;
- hedging transactions;
- special mark-to-market treatment for certain futures positions;
- the “60/40” treatment described for certain noncorporate futures gains/losses.

For study purposes, understand the **conceptual distinction**:

tax treatment depends not only on profit amount, but also on the nature of the position, taxpayer, holding rules, and whether the trade qualifies as a hedge.

Do not treat an old textbook summary as current tax advice. Tax rules can change and depend on jurisdiction.

72. Forward vs futures - the essential comparison

Forward	Futures
Private OTC agreement	Exchange-traded
Customized	Standardized
Often one specified settlement date	Usually contract month / delivery window
Normally settled at maturity	Settled daily through marking to market
Often held to final settlement	Usually closed before delivery
Counterparty credit exposure can be meaningful	Clearing system greatly reduces direct counterparty exposure
Terms negotiated	Terms defined by exchange

Both create an obligation to buy/sell in the future, but the **cash-flow timing and market infrastructure are different**.

73. Forward profit vs futures profit

Suppose a trader is long £1 million at an agreed rate of

$$1.5000 \text{ USD/GBP.}$$

At maturity,

$$S_T = 1.7000.$$

Ignoring the timing of intermediate cash flows, both a corresponding forward and futures position have a total economic gain of

$$1,000,000(1.7000 - 1.5000) = \boxed{\$200,000}.$$

The difference is **when** the money arrives.

Forward:

gain realized at maturity

Futures:

gain/loss realized day by day

This timing can matter because intermediate cash flows can earn interest or require financing.

74. Why daily settlement can make futures and forwards economically different

Suppose a long futures position loses heavily early and recovers later.

The trader must fund early margin calls.

Even if the final total price change is favorable, those early cash outflows had a financing cost.

Conversely, if gains arrive early, the trader may be able to invest them.

Therefore:

Same final price change does not always imply identical present economic value

because futures cash flows occur through time while a forward generally settles at the end.

This idea becomes important in later chapters when comparing forward and futures prices.

75. Foreign-exchange quote inversion

Currency quoting conventions can make two apparently different numbers describe the same exchange rate.

If futures are quoted as

0.9500 USD/CAD,

then the reciprocal quote is

$$\frac{1}{0.9500} = 1.0526 \text{ CAD/USD.}$$

So before comparing two currency quotes, check their orientation.

A useful rule:

$$A/B = x \quad \Rightarrow \quad B/A = \frac{1}{x}$$

Many apparent inconsistencies disappear once you invert one quote.

76. The fastest way to solve a margin problem

When you see a margin question, write these immediately:

$$M_0 = \text{initial margin}$$

M_m = maintenance margin

Q = total underlying units

Then calculate:

$$L = M_0 - M_m$$

where L is the loss capacity before reaching maintenance.

Then:

$$\Delta F = \frac{L}{Q}.$$

Finally decide the direction:

- long position: adverse move = price fall;
- short position: adverse move = price rise.

That solves most Chapter 2 margin-call questions in seconds.

77. The fastest way to solve a futures profit problem

Identify:

- entry futures price F_0 ;
- exit futures price F_1 ;
- contract size Q ;
- number of contracts N ;
- long or short.

Long:

$$\text{Profit} = NQ(F_1 - F_0)$$

Short:

$$\text{Profit} = NQ(F_0 - F_1)$$

If the question asks **when** the profit is realized, remember:

futures gains and losses are realized day by day through marking to market.

78. The fastest way to solve an open-interest problem

Ask what each side is doing.

Buyer	Seller	Effect on OI
Opens new long	Opens new short	+1
Closes short	Closes long	-1
Opens long	Closes long	0
Closes short	Opens short	0

Buyer	Seller	Effect on OI
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Do not assume every trade changes open interest.

Volume always increases when a trade occurs.

Open interest changes only when the number of outstanding contracts changes.

79. The fastest way to identify an order type

Ask what the trader wants to control.

“Execute now.” -> Market order.

“Do not pay more than X / do not sell below X.” -> Limit order.

“If the market moves badly to X, get me out.” -> Stop order.

“If X is hit, activate an order but never execute worse than Y.” -> Stop-limit.

“If the market reaches my favorable target X, execute.” -> Market-if-touched.

Then separately identify the time condition: day, GTC, FOK, etc.

80. The fastest way to analyze delivery questions

Remember the four ideas:

1. most futures are closed out before delivery;
2. the short usually initiates delivery;
3. delivery rules specify asset, grade, place, and time;
4. the possibility of delivery forces convergence.

If a long does not want physical delivery:

close before the relevant notice period begins

according to the specific contract rules.

81. Why speculators are not automatically “bad” for the futures market

A common discussion question argues that futures speculation is pure gambling.

A balanced economic response is:

Speculators do take risks for profit, but their trading can also provide:

- liquidity;
- counterparties for hedgers;
- price discovery;
- narrower bid-ask spreads.

A farmer who wants to short futures to hedge may need someone willing to take the long side.

That counterparty may be a speculator.

The problem is not simply the existence of speculation; the relevant concerns are excessive leverage, manipulation, inadequate controls, and market abuse.

82. Why exchanges care so much about contract design

A successful futures contract needs enough participants on both sides.

Poor design can destroy liquidity.

For example:

- ambiguous asset quality makes longs nervous;
- an enormous contract size excludes smaller users;
- an impractical delivery location raises costs;
- delivery months that do not match commercial needs reduce hedging usefulness;
- weak position rules may invite manipulation.

So contract specification is not administrative trivia. It determines whether the market can function.

83. Why the clearing house does not eliminate all risk

A clearing house reduces direct counterparty risk, but the system still depends on:

- correct margin models;
- liquid markets for closing defaulted positions;
- reliable members;
- sufficient default resources;
- robust operational systems;
- sensible concentration limits.

The clearing house transforms and centralizes risk; it does not make risk disappear.

That is why CCP risk management is itself a major subject in modern derivatives markets.

84. Common Chapter 2 mistakes

Mistake 1: treating margin as the price of the futures contract

Margin is collateral, not the purchase price of the notional exposure.

Mistake 2: thinking maintenance margin is the amount you top back up to

After a margin call, the account is normally restored to the **initial margin** level.

Mistake 3: multiplying open interest by two

Open interest counts contracts, not sides.

Mistake 4: saying volume and open interest are the same

Volume = contracts traded during a period.

Open interest = contracts still outstanding.

Mistake 5: assuming the original counterparty makes delivery to you

The clearing system breaks that direct link.

Mistake 6: assuming stop price is guaranteed execution price

A stop becomes a market order and may execute worse than the trigger level.

Mistake 7: confusing stop order with limit order

Stop is a trigger; limit is a price constraint.

Mistake 8: forgetting unit conversions

Cents per pound, dollars per barrel, number of contracts, and contract multipliers matter.

Mistake 9: assuming futures profit arrives only when you close

Economically the total position may be summarized at close, but cash gains/losses are settled daily.

Mistake 10: comparing currency quotes without checking orientation

USD/CAD and CAD/USD are reciprocals, not directly comparable numbers.

85. Key formulas to memorize

Futures daily gain - long

$$Q(F_t - F_{t-1})$$

Futures daily gain - short

$$Q(F_{t-1} - F_t)$$

Total profit - long

$$NQ(F_{\text{close}} - F_{\text{open}})$$

Total profit - short

$$NQ(F_{\text{open}} - F_{\text{close}})$$

Margin loss capacity

$$M_0 - M_m$$

Approximate adverse price move to maintenance

$$\Delta F = \frac{M_0 - M_m}{NQ}$$

when Q is units per contract and N is number of contracts.

Variation margin required after a call

$$M_0 - \text{current margin balance}$$

Contract count for a simple hedge

$$N = \frac{\text{exposure quantity}}{\text{quantity per contract}}$$

Currency quote inversion

$$x \text{ } A/B \quad \Longleftrightarrow \quad \frac{1}{x} \text{ } B/A$$

86. Verbal answers you should be able to give instantly

What is marking to market?

The daily settlement of gains and losses on a futures position using the exchange's settlement price.

What is initial margin?

Collateral posted when a futures position is opened to support performance of the contract.

What is maintenance margin?

A lower threshold; if the margin balance falls below it, the trader receives a margin call.

What is variation margin?

Additional funds transferred to cover daily losses or restore required margin, depending on the specific margining context.

Why do futures and spot converge?

Because if a material difference remained during delivery, market participants could exploit delivery or arbitrage-like transactions that push the prices back together.

Volume vs open interest?

Volume measures how many contracts traded during the day; open interest measures how many contracts remain outstanding.

How do you close a futures position?

Take the opposite position in the same contract.

Who normally chooses delivery alternatives?

Usually the short, because the short makes delivery, subject to exchange rules.

What is a CCP?

A central counterparty that stands between the two sides of a cleared derivative transaction and manages margin/default risk.

Bilateral vs central clearing?

Bilateral clearing leaves the two parties directly exposed under their agreement; central clearing replaces the bilateral counterparty relationship with exposures to a CCP.

What is a stop order?

An order that becomes a market order after a specified trigger price is reached.

What is a limit order?

An order executable only at the limit price or a more favorable price.

What is a stop-limit order?

An order that becomes a limit order when the stop trigger is reached.

What is open interest?

The number of futures contracts currently outstanding.

What is cash settlement?

Final settlement through a cash payment based on the underlying reference value rather than physical delivery.

Main forward vs futures difference?

Forwards are customized OTC contracts normally settled at maturity; futures are standardized exchange-traded contracts settled daily through a clearing system.

87. Practice-question map: what each group is really testing

The Chapter 2 practice and further questions are not random. They cluster into a few skill families.

Questions on definitions and market structure

Know:

- volume vs open interest;
- FCM vs local;
- bilateral vs central clearing;
- CCP mechanics;
- contract specification;
- delivery rights;
- regulation and manipulation.

Questions on margin

Be able to:

- compute the adverse price move that triggers a margin call;
- determine the variation margin required;
- determine when excess margin may be withdrawn;
- distinguish client margin from clearing-member margin.

Questions on orders

Be able to distinguish:

- market;
- limit;
- stop;
- stop-limit;
- market-if-touched;
- time conditions.

Questions on convergence and arbitrage

Be able to explain:

$$F_T > S_T$$

and

$$F_T < S_T$$

and why the gap cannot persist at delivery.

Questions on profit timing, accounting, and tax

Separate:

1. total economic futures profit;
2. day-by-day cash settlement;
3. accounting recognition;
4. tax recognition.

These are four different questions.

Questions on FX futures and forwards

Check the quote orientation before comparing numbers.

Questions on hedging

Use the sign of the future commercial exposure:

- selling an asset later -> short futures hedge;
 - buying an asset later -> long futures hedge.
-

88. A full mental model for Chapter 2

When somebody enters a futures trade, think through the entire chain:

1. The exchange has already standardized the contract.
2. The trader sends an order through a broker or trading system.
3. A matching opposite order creates a trade.
4. The clearing house becomes central to the obligations.
5. Both sides provide margin.
6. Every day the exchange computes a settlement price.
7. Daily gains and losses move through margin accounts.
8. A losing account can receive a margin call.
9. Most traders close with an offsetting futures trade.
10. If the contract survives into delivery, the exchange's delivery rules apply.
11. The possibility of that delivery forces futures and spot prices to converge.

If this sequence is intuitive to you, Chapter 2 stops feeling like a list of disconnected definitions.

89. What to concentrate on for an internship or oral exam

Be ready to explain, without notes:

1. Why futures need margin even though the initial contract value is approximately zero.
2. How daily settlement works with a numerical example.
3. Initial margin vs maintenance margin vs variation margin.
4. How to calculate a margin-call threshold.
5. Why the clearing house reduces counterparty risk.
6. Why most futures never reach delivery but delivery still determines pricing.
7. Why futures price converges to spot at maturity.
8. Volume vs open interest and how a trade changes open interest.
9. How to read open/high/low/settlement/last/change/volume.
10. Market, limit, stop, stop-limit, and market-if-touched orders.
11. CCP vs bilateral OTC clearing.
12. Collateral, CSA, initial margin, variation margin, and haircuts.
13. Forward vs futures and why daily cash-flow timing matters.
14. Physical delivery vs cash settlement.
15. First notice day, last notice day, and last trading day.
16. The purpose of position limits and anti-manipulation rules.
17. Why speculators can provide useful liquidity to hedgers.
18. Why contract design - quality, size, delivery place, delivery month - matters economically.

If you can explain those topics and perform the margin/profit/open-interest calculations in this guide, you have the core knowledge needed to handle essentially all of Chapter 2's questions.

Final one-page memory sheet

Futures position

Long gains when futures price rises.

Short gains when futures price falls.

Close-out

Long -> sell same contract.

Short -> buy same contract.

Convergence

$$F_T \rightarrow S_T$$

because delivery/arbitrage prevents a persistent maturity gap.

Margin

Initial margin = opening collateral.

Maintenance margin = trigger threshold.

Below maintenance -> margin call.

Top up toward initial margin.

Daily settlement = marking to market.

Profit

Long:

$$NQ(F_1 - F_0)$$

Short:

$$NQ(F_0 - F_1)$$

Open interest

Outstanding contracts, not total sides.

New long + new short -> +1.

Closing long + closing short -> -1.

New + closing -> 0.

Orders

Market = execute now.

Limit = control worst acceptable price.

Stop = trigger market order after adverse level.

Stop-limit = trigger a limit order.

MIT = trigger market order at favorable target.

Clearing

Futures:

Client $\rightarrow$ Broker/Clearing Member $\rightarrow$ Clearing House

OTC central clearing:

$A \rightarrow CCP \rightarrow B$

OTC bilateral clearing:

$A \leftrightarrow B$ with master agreement/CSA/collateral

Forward vs futures

Forward = OTC, customized, terminal settlement.

Futures = standardized, exchange-traded, daily settlement.

That is the mechanical foundation on which the later futures-pricing and hedging chapters are built.